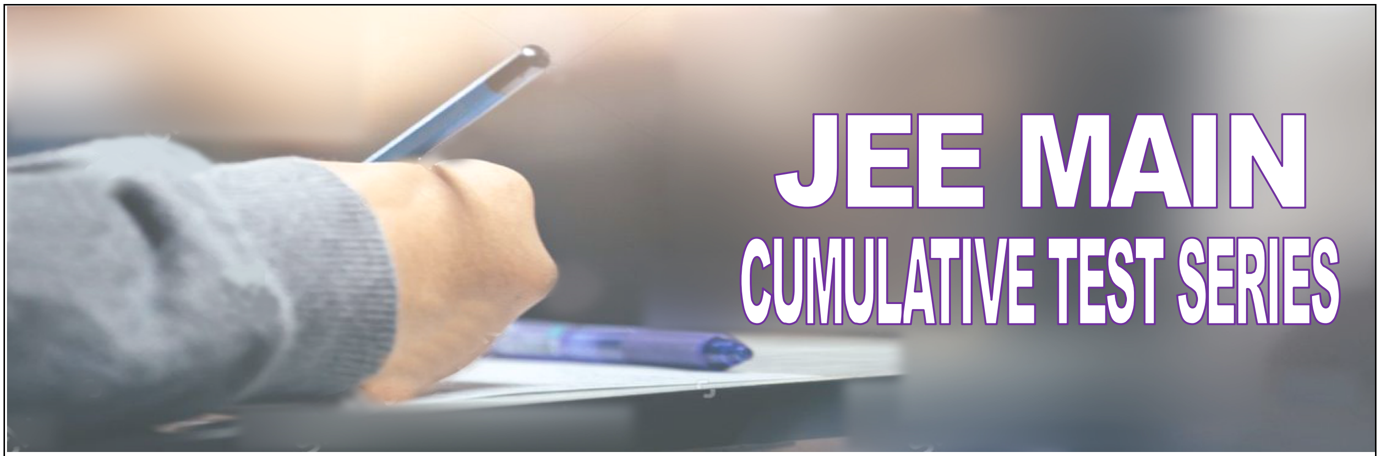


NARAYANA IIT ACADEMY INDIA

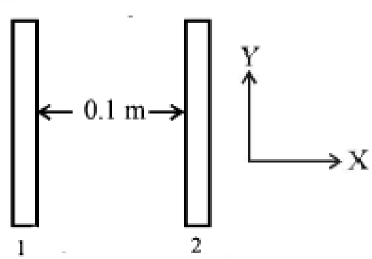
Sec: SR_*CO-SC(MODEL-B) Date:14-07-2024
Time: 3 Hrs CTM-48 Max. Marks: 300

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. The insulating plates are both uniformly charged in such a way that potential difference between them is $V_2 - V_1 = 20V$. (i.e., plate 2 is at higher potential). The plates are separated by d=0.1 m and can be treated as infinitely large. An electron is released from rest on the inner surface of plate 1. What is its speed when it hits plate 2?
$(e = 1.6\times10^{-19} C,\ m_e = 9.11\times10^{-31} kg)$



(1) 2.65×10^{6} m/s (2) 7.02×10^{12} m/s
(3) 1.87×10^{6} m/s (4) 32×10^{-19} m/s

2. When the energy of the incident radiation is increased by 20% the kinetic energy of the photoelectrons emitted from a metal surface increased from 0.5 eV to 0.8 eV. The work function of the metal is:

(1) 0.65Ev (2) 1.0 Ev

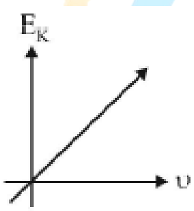
(3) 1.3 Ev (4) 1.5 Ev

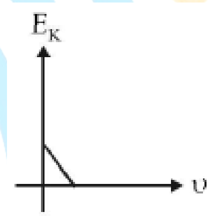
3. The maximum velocity of an electron emitted by light of wavelength λ incident on the surface of a metal of work-function ϕ is

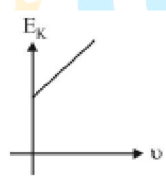
(1) $\sqrt{\dfrac{2(hc+\lambda\phi)}{m\lambda}}$ (2) $\dfrac{2(hc+\lambda\phi)}{m\lambda}$

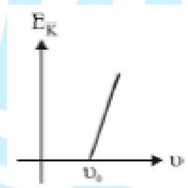
(3) $\sqrt{\dfrac{2(hc+\lambda\phi)}{m\lambda}}$ (4) $\sqrt{\dfrac{2(h\lambda-\phi)}{m}}$

4. Which one of the following graphs represents the variation of maximum kinetic energy (E_K) of the emitted electrons with frequency ν in photoelectric effect correctly?

(1) 

(2) 

(3) 

(4) 

5. If E_2, E_3 are the respective kinetic energies of an electron, an alpha-particle and a proton, each having the same de-Broglie wavelength, then

(1) $E_1 > E_3 > E_2$ (2) $E_2 > E_3 > E_1$

(3) $E_1 > E_2 > E_3$ (4) $E_1 = E_2 = E_3$

6. In photoelectric effect, stopping potential for a light of frequency n_1 is V_1 .if light is replaced by another having a frequency n_2 then its stoping potential will be

(1) $V_1 - \dfrac{h}{e}(n_2 - n_1)$ (2) $V_1 + \dfrac{h}{e}(n_2 + n_1)$

(3) $V_1 + \dfrac{h}{e}(n_2 - 2n_1)$ (4) $V_1 + \dfrac{h}{e}(n_2 - n_1)$

7. A certain metallic surface is illuminated with monochromatic light of wavelength λ. The stopping potential for photo electric current for this light is $3V_0$. If the same surface is illuminated with light of wavelength 2λ, the stopping potential is V_0. The threshold wavelength for this surface for photo-electric effect is

(1) 4λ (2) $\frac{\lambda}{4}$

(3) $\frac{\lambda}{6}$ (3) 6λ

8. The ratio of the respective de Broglie wavelengths associated with electrons accelerated from rest with the voltages 100V, 200V and 300 V is

(1) 1 : 2 : 3 (2) 1 : 4 : 9

(3) $1:\frac{1}{\sqrt{2}}:\frac{1}{\sqrt{3}}$ (4) $1:\frac{1}{2}:\frac{1}{3}$

9. Two radiations of photons energies 1 eV and 2.5 Ev, successively illuminate a photosensitive metalic surface of work function 0.5 eV. The ratio of the maximum speeds of the emitted electrons is :

(1) 1 : 4 (2) 1 : 2

(3) 1: 1 (4) 1 : 5

10. Phoelectric emission is observed from a metallic surface for frequencies $(v_1 > v_2)$. If the maximum values of kinetic energy of the photoelectrons emitted in the two cases are in the ratio of 1 : k, then the threshold frequency of the metallic surface is

(1) $\frac{v_1 - v_2}{k-1}$ (2) $\frac{kv_1 - v_2}{k-1}$

(3) $\frac{kv_2 - v_1}{k-1}$ (4) $\frac{v_2 - v_1}{k}$

11. In the photoelectri ceffect, electrons are emitted

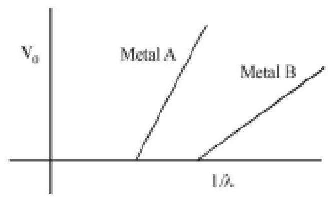
(1) at a rate that is proportiana to the amplitude of the incident radiation

(2) with a maximum velcity proportional to the frequency of the incident radiation

(3) at a rate that is independent of the emitter

(4) only if the frequency of the incident radiations is above a certain thershold value

12. In an experiment on photoelectric effect, a student plots stopping potential V_0 against reciprocal of the wavelength λ of the incident light for two different metals A and B. Thes are shown in the figure.



Looking at the graphs, can most appropriately say that:

(1) Work function of metal B is greater than that of metal A

(2) For light of certain wavelength falling on both metal, maximum kinetic energy of electrons emitted from B

(3) Work function of metal A is greater than that of metal B

(d) student function of metal A is greater than that of metal B

13. Light wavelength 500 mm is incident on a metal with work function 2.288 eV. The Broglie wavelength of the emitted electron is

(1) $\le 2.8\times10^{-9}\,m$ (2) $\ge 2.8\times10^{-9}\,m$

(3) $\le 2.8\times10^{-12}\,m$ (4) $< 2.8\times10^{-10}\,m$

14. An electon of mass m and charge e initially at rest gets accelerated by a constat electic field E. The rate of change of de-Broglic waelength of this electron at time t ignoring relativistic effects is

(1) $\frac{-h}{eEt^2}$ (2) $\frac{-eht}{E}$

(3) $\frac{mh}{eEt^2}$ (4) $\frac{-h}{eE}$

15. The de-Broglie wavelength of neutron in thermal equilibrium at temperature T is

(1) $\frac{30.8}{\sqrt{T}}\overset{0}{A}$ (2) $\frac{3.08}{\sqrt{T}}\overset{0}{A}$

(3) $\frac{0.308}{\sqrt{T}}\overset{0}{A}$ (4) $\frac{0.0308}{\sqrt{T}}\overset{0}{A}$

16. The stopping potential (V_0) versus frequency (v) plot of a substance is shown in figure, the threshold wavelength is

(1) 5×10^{14} m
(2) $6000\overset{0}{A}$
(3) $5000\overset{0}{A}$
(4) Cannot be estimated from given data

17. Surface of certain metal is first illuminated with light of wavelenth $\lambda_1 = 350\text{nm}$ and then, by light of wavelength $\lambda_2 = 540\text{nm}$. It is found that the maximum speed of the photo electron in the two cases differ by a factor of (2) The work function of the metal (in Ev) is close to: (Energy of photon=$\frac{1240}{\lambda(in\ nm)}$eV)

(1) 1.8
(2) 2.5
(3) 5.6
(4) 1.4

18. The magnetic field associated with a light wave is given at the origin by

$B = B_0\left[\sin\left(3.14\times10^7\right)ct + \sin\left(6.28\times10^7\right)ct\right]$.

If this light falls on a silver plate having a work function of 4.7 eV, what will be maximum kinetic energy of the photoelectrons?

(c= $3\times10^8\ \text{ms}^{-1}$, $h = 6.6\times10^{-34}$ J-s)

(1) 6.82 Ev
(2) 12.5 Ev
(3) 8.52 Ev
(4) 7.72 Ev

19. Light wavelength 500 nm falls normal on a slit of width 22.0×10^{-5} cm. The angular position of the second minima from the central maximum will be (in radians)

(1) $\frac{\pi}{8}$
(2) $\frac{\pi}{12}$
(3) $\frac{\pi}{4}$
(4) $\frac{\pi}{6}$

20. A single slit of widdth b is illuminted by a coherent monochromatic light of wavelength λ if the second and fourth minima in the diffraction pattern at a distance 1 m from the slit are at 3 cm and 6 cm respectively from the central maximum, what is the width of the central maximum? (i.e. distance between first minimum on either side of the central maximum)

(1) 1.5 cm
(2) 3.0 cm

(3) 4.5 cm (3) 6.0 cm

SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

21. A beam of plate polarized light of large cross-sectional area and uniform intensity of 3.3 Wm^{-2} falls normally on a polarizer (cross sectional area $3\times10^{-4}m^2$) which rotates about its axis with an angular speed of 31.4 rad/s. The energy of light passing through the polarizer per revolution, is close to:

22. In a double –slit experiment, green light ($5301\overset{0}{A}$) falls on a double slit having a separation of 19.44 μm and a width of 4.05 μm. The number of bright fringes between the first and the second diffraction minima is:

23. Unpolarized light of intesity I passes through and ideal polarizer A. Another identical polarizer B is placed behind A. The intesity of light beyond B is foud to be $\frac{I}{2}$. Now another identical polarizer C is placed between A and B. The intensity beyond B is now found to be $\frac{I}{8}$. The angle between polarizer A and C is:

(1) 0^0 (2) 30^0

(3) 45^0 (4) 60^0

24. In an experiment of single slit diffraction pattern, first minimum for red light concides with first maximum of some other wavelength. If wavelength of red light is $6600\overset{0}{A}$, then wavelength of first maximum will be:

25. A young's double –slit experiment is performed using monocromatic light of wavelength λ. The intensity of light at a point on the screen, where the path difference is λ, is K units. The intensity of light at a point where the path difference if $\frac{\lambda}{6}$ is given by $\frac{nK}{12}$, where n is an integer. The value of n is_____

26. In a Young's double experiment 15 fringes are observed on a small portion of the screen when light of wavelength 500 nm is used. Ten fringes are observed on the same section of the screen when another light source of wavelength λ is used. Then the value of λ is (in nm) _____

27. An Electromagnetic wave of intensity I falls on a surface kept in vaccum at an angle of incidence 60^0.If the surface is 50% reflecting then radiation pressure is $\frac{3I}{KC}$ find 'K'

____

28. In a medium relative permiability μr is 1 and its di electric constant is 9. Then speed of light in that medium is $p\times10^8 m/s$.Then p is ___________

29. Find the angle of deviation for the iocident ray which is initially polarised in the plane of incidence & gets no reflection at the interface of air-glass medium $(\mu_{medium}=\sqrt{3})$

30. The ratio of total energy density (avg) of an EM wave to energy density (avg) due to electric field component in EM wave is x:y (x, y $\rightarrow$primer) then x + y =_____________

CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
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31. Bromide and nitrates both give brown fumes with conc. H_2SO_4. Which of the following reagent will be used to distinguish between them?

A) $K_2Cr_2O_7$ B) $BaCl_2$ C) $FeSO_4$ D) $K_4[Fe(CN)_6]$

32. A solution has a mixture of bromide and iodide salt. When Cl_2 gas is passed followed by the addition of $CHCl_3$, the colour of organic layer will become

A) violet B) reddish brown C) colourless D) blue

33. To identify sulphate, $BaCl_2$ is added. If the salt is silver sulphate, the reagent used to check the sulphate should be

A) $Ba(NO_3)_2$, because all nitrates are soluble in water

B) $BaCl_2$ because $BaSO_4$ formed is insoluble in con. HNO_3

C) $CaCl_2$, because $CaSO_4$ is also water insoluble

D) $CaCl_2$, because $CaSO_4$ is dissolves in con. HNO_3

34. In the ring test of nitrate, a brown coloured ring is formed at the junction of two layers, the ring formed has the following formula.

A) $\left[Fe(H_2O)_5NO\right]SO_4$ with $Fe = +2, NO = 0$

B) $\left[Fe(H_2O)_5NO\right]SO_4$ with $Fe = +3, NO = -1$

C) $\left[Fe(H_2O)_5NO\right]SO_4$ with $Fe = +1, NO = +1$

D) $\left[Fe(H_2O)_5NO\right]SO_4$ with $Fe = 0, NO = +2$

35. From the following information,

$X + H_2SO_4 \longrightarrow Y$ (a colourless and irritating gas)

$Y + K_2Cr_2O_7 + H_2SO_4 \longrightarrow$ (Green coloured solution)

Identify the pair X and Y from the list given below

A) Cl^-, HCl B) S^-, H_2S C) CO_3^{2-}, CO_2 D) SO_3^{2-}, SO_2

36. Chloride salt gives colourless fumes with conc. H_2SO_4, while iodic salt gives violet fumes in the same test. This is because

A) H_2SO_4 reduces HI to I_2 B) the colour of HI is violet

C) HI gets oxidized to I_2 D) HC*l* gets oxidized to Cl_2

37. Sulphide ions form a purple coloured Y compound with X. The oxidation state of central metal in the conversion of X into Y

A) changes from + 2 to +3 B) changes from + 3 to +2

C) changes from + 2 to +4 D) remains the same

38. Mixture of XO_3^{2-} and YO_3^{2-} both are decomposed by dilute H_2SO_4 forming gases which turn lime water milky. If gases are passed first into $K_2Cr_2O_7 / H^+$ solution and then into lime water, then $K_2Cr_2O_7$ solution turned green and lime water milky. Atomic mass of $X > Y$. Thus, mixture contains

A) CO_3^{2-}, SO_3^{2-} B) SO_3^{2-}, CO_3^{2-} C) PO_3^{2-}, CO_3^{2-} D) SO_3^{2-}, PO_3^{2-}

39. A solution of white crystals gives a precipitate with $AgNO_3$ but no precipitate with a solution of Na_2CO_3. The action of conc. H_2SO_4 on the crystals yields a brown gas. The crystals are of

A) $NaNO_3$ B) KCl C) $Ca(NO_3)_2$ D) $NaBr$

40. The acidic solution of a salt produces blue colour with KI starch solution. The salt may be

A) sulphite B) bromide C) nitrite D) chloride

41. A sodium salt of unknown anion when treated with $MgCl_2$ gives white precipitate only on boiling. The anion is

A) SO_4^{2-} B) HCO_3^- C) CO_3^{2-} D) NO_3^-

42. Which of the following gives a suffocating gas when treated with dilute HCl?

A) Carbonate B) Sulphite C) Sulphate D) Borate

43. The bond length in C-O bond in carbon monoxide is 1.128 $\mathring{A}$. The C-O bond in Fe $(CO)_5$ is

A) 1.15 $\mathring{A}$ B) 1.128 $\mathring{A}$ C) 1.072 $\mathring{A}$ D) 1.118 $\mathring{A}$

44. In the isoelectronic series of metal carbonyl, the C-O bond strength is expected to increase in the order.

A) $[Mn(CO)_6]^+ < [Cr(CO)_6] < [V(CO)_6]^-$ B) $[V(CO)_6]^- < [Cr(CO)_6] < [Mn(CO)_6]^+$

C) $[V(CO)_6]^{-1} < [Mn(CO)_6]^+ < [Cr(CO)_6]$ D) $[Cr(CO)_6] < [Mn(CO)_6]^+ < [V(CO)_6]^-$

45. Which one of the following statements is incorrect?

A) Greater the stability constant of a complex ion, greater is its stability

B) Greater the charge on the central metal ion, greater is the stability of the complex

C) Greater the basic character of the ligand, the great is the stability of complex

D) Complexes have low stability constants

46. Synergic bonding in metal carbonyl complexes [M=Metal]

i) Decreases $C-O$ bond strength

ii) Increases $M-C$ bond length

iii) increases bond order for $M-C$ bond

iv) increases paramagnetism of complex

A) i, ii B) i, iii C) i, ii, iv D) ii, iv

47. $S_2O_3^{2-} \xrightarrow{Ag^+} \underset{(\text{Clear solutiomn})}{X} \xrightarrow{Ag^+} \underset{(\text{white ppt})}{Y} \xrightarrow{\text{with time}} \underset{(\text{black ppt})}{Z}$ In the above reaction sequence in aqueous solution the species x, y and z respectively are

A) $\left[Ag(S_2O_3)_2\right]^{3-}, Ag_2S_2O_3, Ag_2S$ B) $\left[Ag(S_2O_3)_2\right]^{5-}, Ag_2S_2O_3, Ag_2S$

C) $\left[Ag(SO_3)_2\right]^{3-}, Ag_2S_2O_3, Ag$ D) $\left[Ag(SO_3)_3\right]^{3-}, Ag_2S_2O_4, Ag$

48. In the borax-bead test a colourless become coloured on heating with a transition metal salt. This is due to the formation of coloured

A) Boric acid B) Borate and metaborate of the metal

C) orthoborate of the metal D) Hexaborate of the metal

49. The Brown ring test is not reliable in the presence of

A) Br^- ions B) SO_4^{-2} ions C) Zn^{+2} ions D) Al^{+3} ions

50. Sulphite on treatment with dilute H_2SO_4 liberates a gas which

A) Turns lead acetate paper black B) Turns acidified $K_2Cr_2O_7$ paper green

C) Smells like vinegar D) Burns with blue flame

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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51. $NaCl + K_2Cr_2O_7 + conc.H_2SO_4 \xrightarrow{\Delta} \text{Orange red gas} \xrightarrow[\text{NaOH sol.}]{\text{dilute}} \text{yellow solution} \xrightarrow[Pb(CH_3COO)_2]{CH_3COOH} \underset{(\text{Yellow ppt})}{X}$

What is the number of oxygen atoms in one molecule of X?

52. $Fe_4\left[Fe(CN)_6\right]_3$ (Prussian blue) dissolves in excess of $K_4\left[Fe(CN)_6\right]$ forming $KFe\left[Fe(CN)_6\right]$. Sum of oxidation number of iron atoms in it is.

53. How many of the following liberate gases with dilute or conc H_2SO_4.

$Na_2S, Na_2SO_3, NaCl, NaBr, KI, KNO_2, KNO_3, Na_2SO_4, Na_2CO_3$

54. How many of the following do not give chromyl chloride test.

$PbCl_2, HgCl_2, AgCl, SnCl_4, Hg_2Cl_2, NaCl, SbCl_3, CuCl_2$

55. A deep brownish red colour gas is formed, when a mixture of KCl and $K_2Cr_2O_7$ is heated with conc. H_2SO_4. What is the oxidation number of the metal in this coloured gas?

56. In how many of the following combinations coloured gases will be liberated

i) $Na_2S + \text{dil } HCl$ ii) $NaCl + \text{dil } HCl$ iii) $KBr + \text{conc. } H_2SO_4$

iv) $NaNO_3 + Cu + \text{conc.} H_2SO_4$ v) $Na_2SO_3 + \text{dil } HCl$

vi) $NaCl + \text{conc } H_2SO_4 + MnO_2$ vii) $KI + \text{conc.} H_2SO_4$

57. The number of terminal carbonyl groups present in $Fe_2(CO)_9$ are

58. $EDTA^{4-}$ is ethylenediaminetetraacetate ion. The total number of $N-Co-O$ bond angles in $[Co(EDATA)]^{1-}$ complex ion is

59. How many of the following anions is give precipitate with silver nitrate solution.

KI , Na_2S , Na_2SO_3 , KNO_2 , $NaCl$, KNO_3 , Na_2CO_3 , $NaBr$

60. How many of the following give a colored bead in borax bead test

$Cu(NO_3)_2, FeSO_4, CrCl_3, ZnSO_4, NiSO_4, CoCl_2, AlCl_3$

MATHEMATICS MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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61. 6 Married couples are standing in a room. If 4 people are chosen at random, then the chance that exactly one married couple is among the 4 is

A) $\frac{16}{33}$ B) $\frac{8}{33}$ C) $\frac{17}{33}$ D) $\frac{24}{33}$

62. A quadratic equation is chosen from the set of all the quadratic equations which are unchanged by squaring their roots. The chance that the chosen equation has equal roots is

A) $\frac{1}{2}$ B) $\frac{1}{3}$ C) $\frac{1}{4}$ D) $\frac{2}{3}$

63. A cube with all six faces coloured is cut into 64 cubical blocks of the same size which are thoroughly mixed. Find the probability that the 2 randomly chosen block have 2 coloured faces each

A) $\frac{23}{268}$ B) $\frac{23}{216}$ C) $\frac{23}{168}$ D) None

64. Consider a function f(x) that has zeroes 4 and 9. Given that Mr.A randomly selects number from the set{-10,-9,-8……….8,9,10}, what is the probability that Mr. A chooses a zero of $f(x^2)$?

A) $\frac{6}{21}$ B) $\frac{4}{21}$ C) $\frac{7}{21}$ D) $\frac{8}{21}$

65. Mr. A lives at origin on the cartesian plane and has his office at (4,5). His friend lives at (2,3) on the same plane Mr. A can go his office travelling one block at time either in the +y or +x direction. If all possible paths are equally likely then the probability that Mr. A passed his friends house is (Shortest path for any event must be considered)

A) $\frac{1}{2}$ B) $\frac{10}{21}$ C) $\frac{1}{4}$ D) $\frac{11}{21}$

66. I have 3 normal dice, one red, one blue and one green and I roll all three simultaneously Let P be the probability that the sum of the numbers on the red and blue dice is equal to

A) 79 B) 77 C)61 D)57

67. There are three passengers on airport shuttle bus that makes stops at four different hotel. The probability that all three passengers will be staying at different hotels is

A) $\frac{1}{16}$ B) $\frac{1}{4}$ C) $\frac{3}{8}$ D) $\frac{3}{4}$

68. Of all the mappings that can be defined from the set $A:\{1,2,3,4\} \rightarrow B:\{5,6,7,8,9\}$, a mapping is randomly selected. The chance that the selected mapping is strictly monotonic is

A) $\frac{1}{125}$ B) $\frac{2}{125}$ C) $\frac{5}{4096}$ D) $\frac{5}{2048}$

69. Letter of the word TITANIC are arranged to form all the possible words. What is the probability that a word formed starts either with a T or vowel?

A) $\frac{2}{7}$ B) $\frac{4}{7}$ C) $\frac{3}{7}$ D) $\frac{5}{7}$

70. A 4-digit number is randomly picked from all the 4 digits' numbers, then the probability than the product of its digit is divisible by 3 is

A) $\frac{107}{125}$ B) $\frac{109}{125}$ C) $\frac{111}{125}$ D) None of these

71. Three numbers are randomly selected from the set {10,11,12……….,100}. Probability that they form a Geometric progression with integral common ratio greater than 1 is:

A) $\frac{15}{{}^{91}C_3}$ B) $\frac{16}{{}^{91}C_3}$ C) $\frac{17}{{}^{91}C_3}$ D) $\frac{18}{{}^{91}C_3}$

72. A seven digit number of the from $a_1a_2a_3a_4a_5a_6a_7$ (all digits different) randomly formed then the probability that this will satisfies condition $a_1>a_2>a_3>a_4<a_5<a_6<a_7$ is______

A) $\frac{5}{1134}$ B) $\frac{7}{1134}$ C) $\frac{9}{1134}$ D) $\frac{5}{648}$

73. A letter is choosen at random out of 'ASSININE' and one is chosen at random out of 'ASSASSIN'. Find the probability that the same letter is chosen on both occasions.

A) $\frac{7}{16}$ B) $\frac{7}{32}$ C) $\frac{5}{16}$ D) $\frac{9}{32}$

74. The number of ways in which 21 identical apples can be distributed among three children such that each child gets at least 2 apples is

A) 406 B)130 C) 142 D)136

75. The number of ways to distribute 30 identical candies among four children C_1, C_2, C_3, C_4 so that C_2 receives atleast 4 and atmost 7 candies, C_3 receives atleast 2 and atmost 6 candies, is equal to

A) 205 B) 615 C)510 D)430

76. The distance of the point $(-1,9,-16)$ from the plane $2x+3y-z=5$ measured parallel to the line $\frac{x+4}{3}=\frac{2-y}{4}=\frac{z-3}{12}$ is

(A) $13\sqrt{2}$ (B) 31 (C) 26 (D) $20\sqrt{2}$

77. If $2\tan^2\theta-5\sec\theta=1$ has exactly 7 solutions in the interval $\left[0,\frac{n\pi}{2}\right]$, for the least value of $n\in N$ then $\sum_{k=1}^{n}\frac{k}{2^k}$ is equal to :

(A) $\frac{1}{2^{15}}\left(2^{14}-14\right)$ (B) $\frac{1}{2^{14}}\left(2^{15}-15\right)$ (C) $1-\frac{15}{2^{13}}$ (D) $\frac{1}{2^{13}}\left(2^{14}-15\right)$

78. Let S= {1,2,3,4,5,6}. Then the probability that a randomly chosen onto function g from S to S satisfies. g(3)=2g(1):

A) $\frac{1}{10}$ B) $\frac{1}{15}$ C) $\frac{1}{5}$ D) $\frac{1}{30}$

79. Let $a_1=1, a_2, a_3, a_4, ...$ be consecutive natural numbers. Then

$$\tan^{-1}\left(\frac{1}{1+a_1a_2}\right)+\tan^{-1}\left(\frac{1}{1+a_2a_3}\right)+......+\tan^{-1}\left(\frac{1}{1+a_{2021}a_{2022}}\right)$$ is equal to

(A) $\frac{\pi}{4}-\cot^{-1}(2021)$ (B) $\cot^{-1}(2022)-\frac{\pi}{4}$ (C) $\tan^{-1}(2022)-\frac{\pi}{4}$ (D) $\frac{\pi}{4}-\tan^{-1}(2022)$

80. Let $\alpha\in(0,\infty)$ and $A=\begin{bmatrix}1&2&\alpha\\1&0&1\\0&1&2\end{bmatrix}$. If $\det\left(adj\left(2A-A^T\right).adj\left(A-2A^T\right)\right)=2^8$, then $\left(\det(A)\right)^2$ is equal to:

(A) 36 (B) 16 (C) 1 (D) 49

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.s

81. Let the line of the shortest distance between the lines

$$L_1:\vec{r}=\left(\hat{i}+2\hat{j}+3\hat{k}\right)+\lambda\left(\hat{i}-\hat{j}+\hat{k}\right) and$$

$$L_2:\vec{r}=\left(4\hat{i}+5\hat{j}+6\hat{k}\right)+\mu\left(\hat{i}+\hat{j}-\hat{k}\right)$$

Intersect L_1 and L_2 at P and Q respectively. If (α,β,γ) is the midpoint of the line segment PQ, then $2(\alpha+\beta+\gamma)$ is equal to

82. A man sent 7 letters to his 7 friends. The letters are kept in the addressed envelopes at random. The probability that 3 friends receive correct letters and 4 letters go to wrong destinations is equal to $\frac{a}{b}$, then a+b is _______

83. Let N denote the number that turns up when a fair die is rolled. If the probability that the system of equations x+y+z=1, 2x+Ny+2z=2, 3x+3y+Nz=3 has unique solution is $\frac{k}{3}$ then K=_____

84. Let $f(x)=x^3+ax^2+bx+c$, where a,b,c are chosen respectively by throwing a die there times. The probability that f(x) is an strictly increasing function $\forall x \in R$ is equal to $\frac{a}{b}$ where a,b are co-prime natural numbers, then b-a is equal to

85. Let A={1,2,3,5,8,9}. then the number of possible functions f: $A \rightarrow A$ such that f(m.n)=f(m).f(n) for every m,n, $\in A$ with m x n, $\in A$ is equal to ___________.

86. Let S={1,2,3,4,5,6,9}. Then the number of elements in the set $T=\{A \subseteq S : A \neq \phi$ and the sum of all the elements of A is not a multiple of 3} is.

87. Let A={0,1,2,3,4,5,6,7}. Then the number of bijective function $f:A \rightarrow A$ such that $f(1)+f(2)=3-f(3)$ is equal to

88. Three a's, three b's and three c's are placed randomly in a 3×3 matrix. The probability that no row or column contain two identical letters can be expressed as $\frac{p}{q}$, where p and q are co-prime then (p+q) equals to:

89. Three different numbers are selected at random from the set A={1,2,3,……10}. Then the probability that the product of two numbers equal to the third number is $\frac{p}{q}$, where p and q are relatively prime positive integers then the value of (p+q) is:

90. The number of solutions of $\sin^2 x+\left(2+2x-x^2\right)\sin x-3(x-1)^2=0$, where $-\pi \leq x \leq \pi$, is _____